

Vietnam's Risk of the Middle Income Trap: The Role of Child-Rearing Costs

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1 Introduction

The [World Bank \(2025\)](#)'s "Viet Nam 2045—Breaking Through: Institutions for a High-Income Future" report highlights the looming risk of the middle income trap to Vietnam. The term middle income trap refers to the empirical regularity that economies which successfully transition from low- to middle-income status frequently fail to make the subsequent transition to high-income status ([Eichengreen, Park, & Shin, 2014](#); [Agénor, 2017](#)). Hence, a trap. Furthermore, recent commitments of the Vietnam's authority of 10% GDP growth rate in 2026, upper-middle-income status by 2030, and high-income status by 2045, institutionalized in Resolution No.25/2026/QH16 issued by the National Assembly, show significant endeavor to advance the country away from the middle income status.

However, as the [World Bank \(2025\)](#) focuses on institutional reforms, this policy note will complement their approach by addressing the quantity-quality tradeoff regarding child-rearing costs. In this case, we consider Vietnam's narrow window to escape the middle income trap, regarding its current young working population advantage, i.e., population dividend, to mitigate the risk of "getting old before getting rich." This policy note will proceed as follows: first, we will consider empirical facts about the Vietnam economy; then discussing the theoretical overlapping generations model with endogenous population growth rate, as well as simulation results; and finally, policy recommendations for Vietnam's authority. The cornerstone of this policy note is describing a theoretical threshold, dictated by low-high wage regimes, which divides the Vietnam economy's growth paths into low income and high income scenarios. The key approach to surpass this threshold is investing in human capital and maintaining sustainable fertility rate by making policy interventions on child-rearing quantity-quality tradeoff.

2 Stylized Facts

Regarding Vietnam's economic growth, we present three stylized facts:

1. First, Vietnam's productivity engine is stalling relative to the Asian Miracles benchmark. Considering Total Factor Productivity (TFP), the residual growth unexplained by capital and labor accumulation, has contributed approximately 35% of Vietnam's GDP growth in 2011–2020, down from its 1990s peak and well below the 50% share sustained by Korea and Singapore during their own escape from the middle income trap (see [Figure 1](#) for Vietnam and its peer TFP index paths, normalized to 1.0 in 2000).
2. Second, Vietnam exports more, but not smarter. Its high-tech share has not tracked income growth the way Korea's did at comparable stages (see [Figure 2](#)).
3. Third, Vietnam's demographic window is closing (see [Figure 3](#)), indicating that quantity (labor

force side and golden population) can no longer substitute for quality (education and human capital).

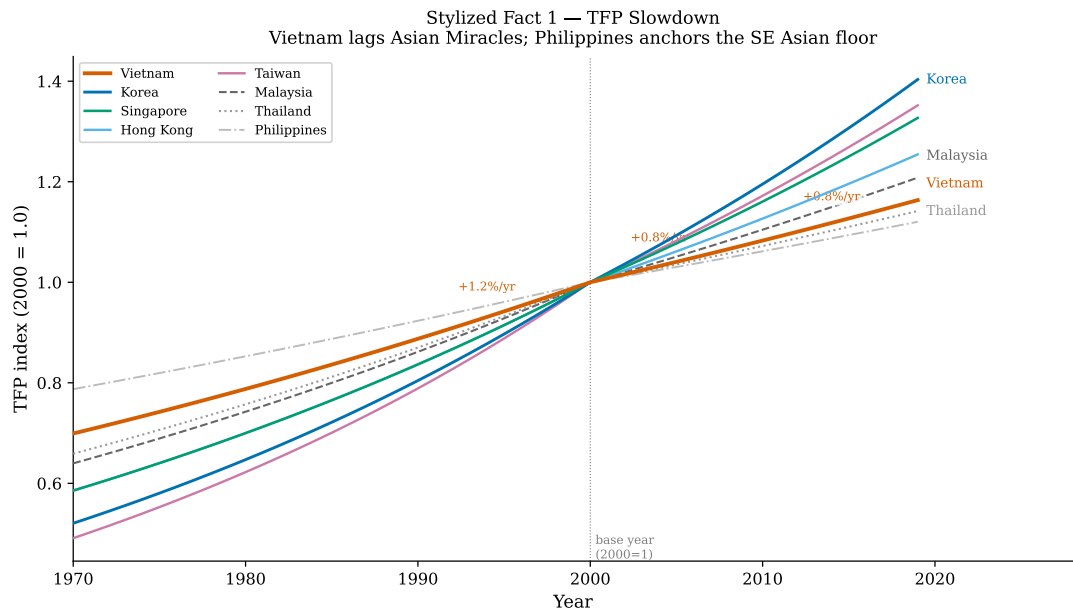


Figure 1: Vietnam's Total Factor Productivity versus neighboring countries
Source: Penn World Tables 10.01 (Feenstra, Inklaar, & Timmer, 2015), variable `rtfpna` (TFP at constant national prices, 2017 = 1), normalised to 1.0 in 2000. Asian Miracles: Korea, Singapore, Hong Kong, Taiwan. SE Asian peers: Malaysia, Thailand, Philippines.

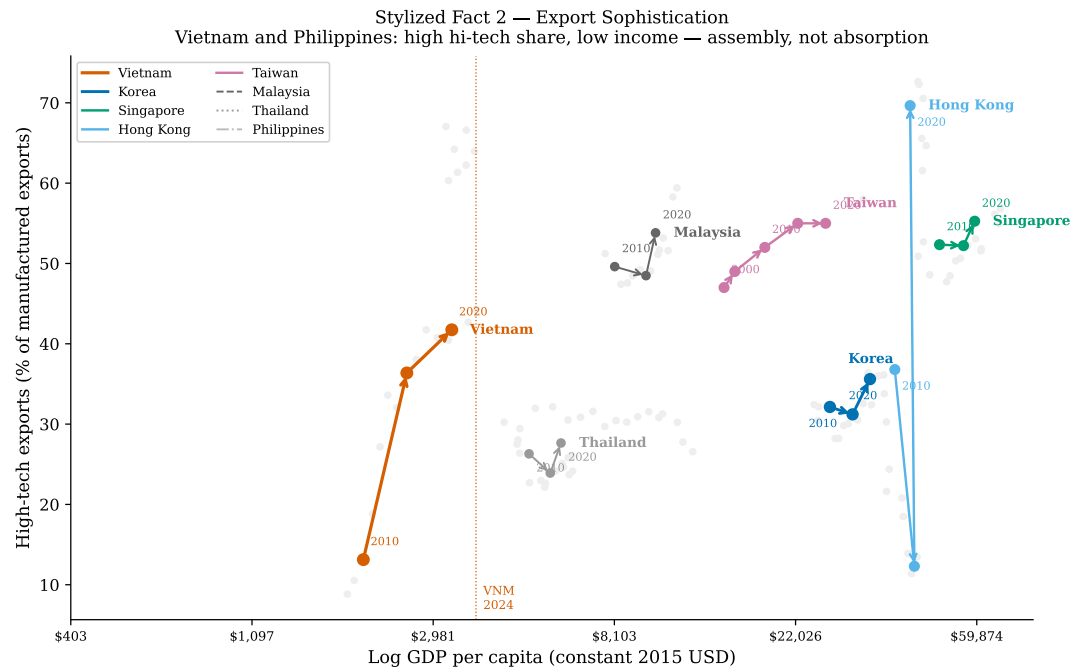


Figure 2: Vietnam's Exports versus neighboring countries
Source: World Bank World Development Indicators (WDI). High-technology exports as a share of manufactured exports (`TX.VAL.TECH.MF.ZS`) and GDP per capita in constant 2015 USD (`NY.GDP.PCAP.KD`). Trajectories shown at five-year intervals, 2000–2020.

Stylized Fact 3 — Fertility Collapse
Asian Miracles fell earliest; Vietnam and Philippines still diverge

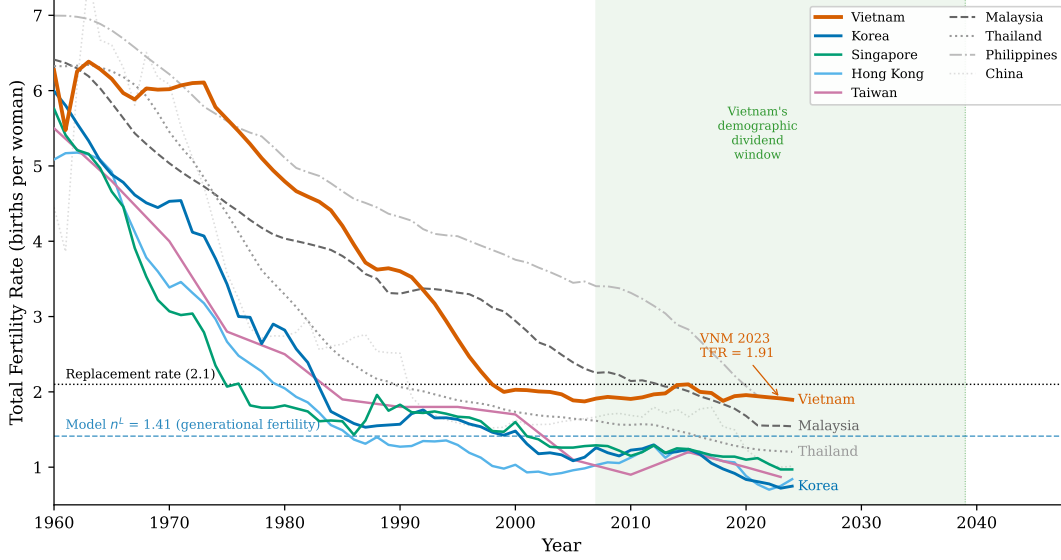


Figure 3: Vietnam's Total Fertility Rate versus neighboring countries

Source: World Bank World Development Indicators (WDI), SP.DYN.TFRT.IN (1960–2023). Taiwan from National Development Council (NDC Taiwan). Replacement rate at TFR = 2.1. Model threshold $n^L = 1.41$ from calibrated OLG parameters.

3 Theoretical Model

3.1 Modeling

We consider an Overlapping Generations (OLG) model of Vietnam's economy with endogenous population growth, following the theoretical fashion of [Becker and Lewis \(1973\)](#) and [Galor and Weil \(2000\)](#), where households choose simultaneously how many children to have (n_t) and how much to invest in each child's human capital (e_t), subject to a time-cost of child-rearing. At low wages, the cost of children is low and households optimally choose high quantity, low quality. At high wages, the opportunity cost of child-rearing rises and households substitute toward quality.

The model is set up the same as the conventional OLG model, time is discrete, $t = 0, 1, 2, \dots$. Each generation of household lives two periods, old and young. At date t there are two sets of households, N_t young household and N_{t-1} old household. We model the population law of motion as

$$N_{t+1} = n_t N_t \quad (1)$$

with endogenous replacement rate n_t as the average number of children per young household, optimally chosen at date t .

The young household seeks to maximize its lifetime utility:

$$\max_{c_t^y, c_{t+1}^o, s_t, n_t, e_t} \ln(c_t^y) + \beta \ln(c_{t+1}^o) + \gamma \ln(n_t h(e_t)) \quad (2)$$

subject to

$$c_t^y + s_t + n_t e_t = (1 - \rho n_t) w_t \quad (3)$$

$$c_{t+1}^o = \tilde{R}_{t+1} s_t \quad (4)$$

where c_t^y, c_{t+1}^o are young and old consumption, n_t is the number of children, h_{t+1} is the human capital each child will carry into adulthood, $\gamma > 0$ represents shadow cost of offspring, and $\tilde{R}_{t+1} \equiv R_{t+1} + (1 - \delta)$ is the gross return on savings.

The human capital of each child, as a function of parental investment e_t , is:

$$h_{t+1} = h(e_t) = (\underline{e} + e_t)^\mu \quad (5)$$

where parameter $\mu \in (0, 1)$ dictates the curvature of the human capital production function, and $\underline{e} > 0$ is the baseline human capital endowment that obtainable even if the parents invest nothing. We assume two properties hold. First, $h(0) = \underline{e}^\mu > 0$ indicates that even with no parental investment, children still have positive human capital. Second, $h'(e_t) = \mu(\underline{e} + e_t)^{\mu-1}$ means the marginal return to education is decreasing, but bounded above by $\mu\underline{e}^{\mu-1}$ at $e_t = 0$.

The young household's optimal solutions are:

$$\frac{c_t^{y*}}{w_t} = \frac{1}{1 + \beta + \gamma} \quad (6)$$

$$\frac{s_t^*}{w_t} = \frac{\beta}{1 + \beta + \gamma} \quad (7)$$

$$\frac{n_t^* e_t^* + \rho n_t^* w_t}{w_t} = \frac{\gamma}{1 + \beta + \gamma} \quad (8)$$

where we notice that the young household has to allocate the child-rearing budget between quantity n_t and quality e_t (Equation 8), regarding fraction $\gamma/(1 + \beta + \gamma)$ of labor income w_t . Holding the total child-rearing budget $Z_t \equiv \frac{\gamma w_t}{1 + \beta + \gamma}$ fixed, the household solves:

$$\max_{n_t, e_t} \ln(n_t) + \mu \ln(\underline{e} + e_t) \quad (9)$$

subject to:

$$n_t e_t + \rho n_t w_t = \frac{\gamma}{1 + \beta + \gamma} w_t \quad (10)$$

that results in the optimal solution:

$$e_t^* = \frac{\mu \rho w_t - \underline{e}}{1 - \mu} \quad (11)$$

which is the interior optimum, occurring when $w_t > \hat{w} \equiv \frac{\underline{e}}{\mu \rho}$, where $\hat{w}$ is the critical wage defining two regimes:

1. Regime L (low wage, $w_t \leq \hat{w}$). Household invest nothing in child quality e_t beyond the baseline $\underline{e}$, and the entire child-rearing budget goes to quantity:

$$e_t = 0, \quad n_t^L = \frac{Z_t}{\rho w_t} = \frac{\gamma}{(1 + \beta + \gamma)\rho} \equiv n^L \quad (12)$$

2. Regime H (high wage, $w_t > \hat{w}$). Household invest a positive amount per child:

$$e_t^* = \frac{\mu \rho w_t - \underline{e}}{1 - \mu}, \quad n_t^H = \frac{Z_t}{e_t^* + \rho w_t} = \frac{\gamma w_t (1 - \mu)}{(1 + \beta + \gamma)(\rho w_t - \underline{e})} \equiv n^H(w_t) \quad (13)$$

Regarding capital accumulation, we have aggregate savings of the young at date t become aggregate capital at date $t + 1$:

$$K_{t+1} = s_t^* N_t = \frac{\beta}{1 + \beta + \gamma} w_t N_t = \frac{\beta(1 - \alpha) A k_t^\alpha h_t N_t}{1 + \beta + \gamma} \quad (14)$$

where we define $k_{t+1} \equiv \frac{K_{t+1}}{h_{t+1} N_{t+1}}$ and $\Phi \equiv \frac{\beta(1 - \alpha) A}{1 + \beta + \gamma}$ to obtain the capital per effective worker law of motion:

$$k_{t+1} = \frac{1}{n_t} \frac{h_t}{h_{t+1}} \Phi k_t^\alpha \quad (15)$$

so we can obtain two steady states. First, for Regime L, $n_t = n^L$ and $h_t = h_{t+1} = \underline{e}^\mu$ (constant), so $k_{t+1} = (\Phi/n^L) k_t^\alpha$, giving:

$$k_L^* = \left[\frac{\Phi}{n^L} \right]^{\frac{1}{1 - \alpha}} = \left[\frac{\beta(1 - \alpha) A \rho}{\gamma} \right]^{\frac{1}{1 - \alpha}} \quad (16)$$

Second, for Regime H, at steady state $h_t = h_{t+1}$ so the h_t/h_{t+1} ratio drops out, giving $(k_H^*)^{1 - \alpha} = \Phi/n^H$. In the asymptotic case $\rho w_H^* \gg \underline{e}$, fertility simplifies to $n^H \approx \gamma(1 - \mu)/[(1 + \beta + \gamma)\rho] = n^L(1 - \mu)$, and therefore:

$$k_H^* = \left[\frac{\Phi}{n^L(1 - \mu)} \right]^{\frac{1}{1 - \alpha}} = \left[\frac{\beta(1 - \alpha) A \rho}{\gamma(1 - \mu)} \right]^{\frac{1}{1 - \alpha}} \quad (17)$$

3.2 The Middle Income Trap

Given income per effective working household:

$$y_t \equiv \frac{Y_t}{h_t N_t} = \frac{A K_t^\alpha (h_t N_t)^{1 - \alpha}}{h_t N_t} = A k_t^\alpha \quad (18)$$

where there is the steady state structure:

$$k_L^* < \hat{k}_M < k_H^* \quad (19)$$

we also have the income per effective working household structure:

$$y_L^* < \hat{y}_M < y_H^* \quad (20)$$

where the per effective worker income gap between regimes is

$$\frac{y_H^*}{y_L^*} \equiv \frac{(Y/N)_H^*}{(Y/N)_L^*} = \left(\frac{k_H^*}{k_L^*} \right)^\alpha \frac{h_H^*}{\underline{e}^\mu} = (1 - \mu)^{-\alpha/(1 - \alpha)} \frac{h_H^*}{\underline{e}^\mu} > 1 \quad (21)$$

There are two illustrations of the middle income trap, dictated by the switching between Regime L and Regime H. Regarding capital accumulation structure (Equation 19), Figure 4 shows two scenarios—a low capital level k_L^* and a high capital level k_H^* (varying by μ). Figure 5 accommodates the

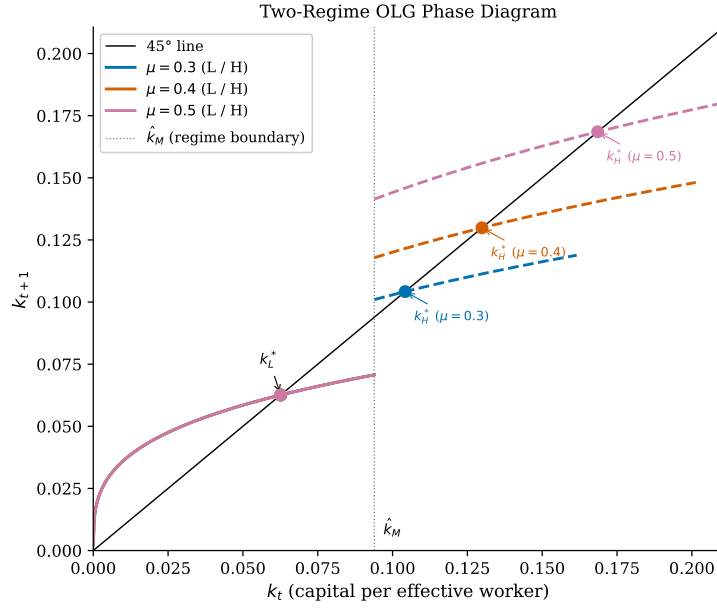


Figure 4: Steady State Capital Accumulation in Two Regimes
 Calibrated to Vietnam macro data (World Bank & Penn World Table, 1986–2024). One OLG period equals one generation ($T = 25$ years). Parameters are $\alpha = 0.30$, $A = 1.0$, $\delta = 0.05$, $\rho = 0.08$, $\beta = 0.486$, $\gamma = 0.189$.

income structure (Equation 20), indicating that Vietnam could transition to a higher level of income $\tilde{y}_H^*$ from a lower level of income $\tilde{y}_L^*$ if the country can surpass the $\hat{y}_M$ threshold. This is the cornerstone of the analysis, where we can focus on policy recommendations for how Vietnam can exceed the $\hat{y}_M$ threshold and escape the middle income trap. Furthermore, the role of μ as the productivity of schooling, capturing the return on education investment regarding how effectively Vietnam's schooling system turns spending on children into skills that raise their adult wages, also determines which high-income status is within reach.

4 Policy Recommendations

This policy note proposes two main approaches to address the middle income trap regarding household's decision in human capital development and child-rearing cost. The model delivers a verdict that Vietnam's escape from the trap is not a savings problem but a child quantity-quality tradeoff. In this case, Vietnamese households are not spending enough on each child's education and development relative to what it takes to push the economy onto a sustained high-income path. Thus, Vietnamese policymakers can consider the following two routes.

Subsidising education to effectively lower the relative price of child quality. The central idea is that when the government lowers the private cost of investing in a child's education, households voluntarily shift their spending toward quality. Thus, it is not solely a welfare transfer but a change in structural optimization that drives the quality-quantity trade-off at the household level.

Fertility policy or child-rearing cost reduction. The central idea is that raising a child in Vietnam is expensive in two distinct ways: it costs money (fees, materials, food, education) and it costs time (foregone work hours for the parent providing care). These costs create a tradeoff between having more children versus investing more in each one. By reducing the cost, households will automatically

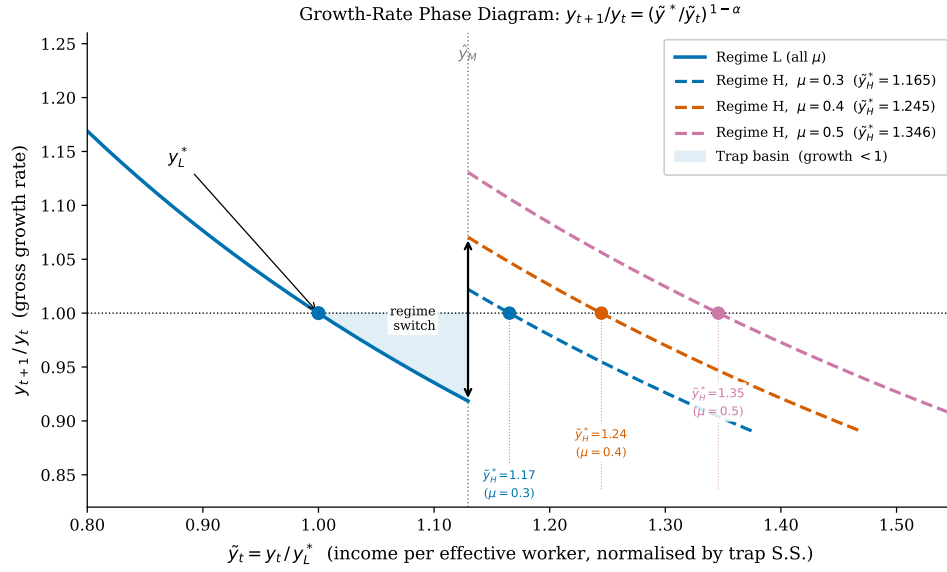


Figure 5: Steady State Income per Effective Worker in Two Regimes
 Calibrated to Vietnam macro data (World Bank & Penn World Table, 1986–2024). One OLG period equals one generation ($T = 25$ years). Parameters are $\alpha = 0.30$, $A = 1.0$, $\delta = 0.05$, $\rho = 0.08$, $\beta = 0.486$, $\gamma = 0.189$.

shift their child-bearing budget toward higher quality investment. Making children cheaper to have can pull families toward having more children at lower quality per child.

There are signs that the Vietnam's authority has already taken these notions into account. In September 2025, the Government announced the No. 238/2025/ND-CP Decree to waive tuition fees for students in public schools for all grades. In February 2025, the No. 219/QD-TT Decision of the Prime Minister on approving the “Program to Ensure Replacement-Level Fertility Nationwide by 2030” shows policy action on addressing fertility rate. However, there are limited empirical evidence of their effectiveness on Vietnam's economic growth.

In sum, this policy note has discussed the theoretical risk of the middle income trap for the case of Vietnam. We utilized the OLG with endogenous population growth rate model, where households optimally decide the number of children and the amount to invest in their offspring. According to the model, Vietnam must surpass the threshold $(\hat{y}_M, \hat{k}_M)$ to successfully escape the middle income trap. Thus, there are two policy routes that are recommended, and there are positive signals that Vietnamese policymakers are aware of this issue, regardless implementation capacity and policy effectiveness. This policy note concludes that Vietnam's quest for growth, especially the aspiring 10% GDP growth rate this year, 2026, are not entirely unapproachable, but will need significant structural considerations on Vietnam's macroeconomic outlook and its demography.

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